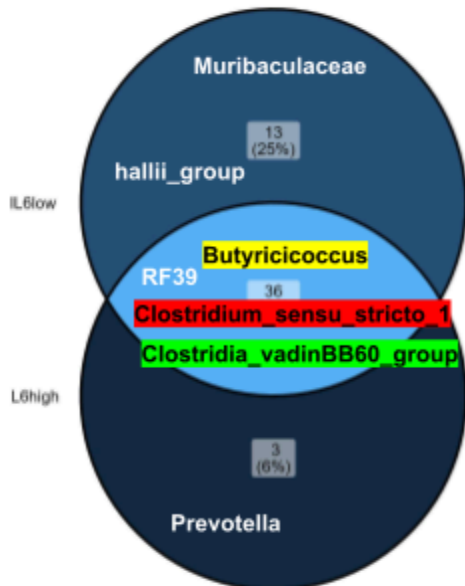
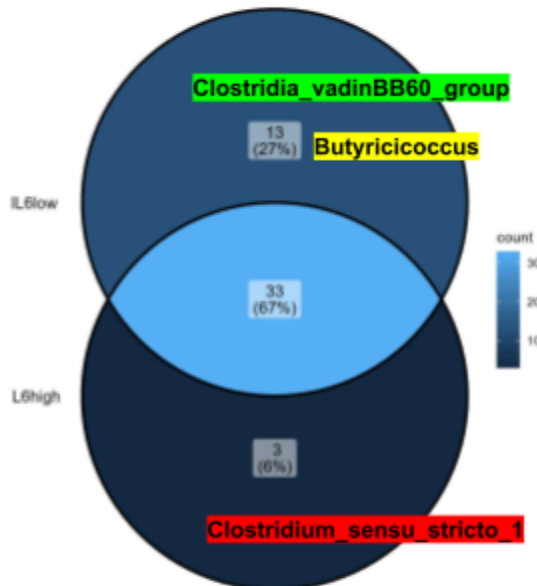


Core ASVs in IL6 High vs Low groups
(70% prevalence), all patients, Genus Glommed



All patients

Core ASVs in IL6 High vs Low groups
(70% prevalence), HIV+ only, Genus Glommed



HIV+ only

Method: Sorted genres produced by core microbiome analysis were put into a CSV and manually inspected for changes.

- COLOR HIGHLIGHTS denote genres which are found in one category within a certain HIV status, but in another when you consider another HIV status.
- WHITE TEXT denote genres found exclusively in that condition.
- ALL OTHER GENUSES were in the same position.

Found in IL6 high of BOTH:

g__Anaerostipes

- Butyrate producing species
 - Essential for maintaining intestinal epithelial cell barrier fxn
 - Regulates immune response of intestinal mucosa
 - Implicated in an anti-cancer role.
- Decrease in this species associated with...
 - Ulcerative colitis
 - Crohn's disease
 - Intestinal Lymphoma
 - Type 2 diabetes
- Not ALL species in this genus have necessarily positive effects on host
- From PMID: 38470044

g__CAG-56

- Not well studied

OTHERS, AS DENOTED BY CHART ABOVE:

Butyricicoccus**

- Reduced abundance of genes of this genus associated with human intestinal disorders: including IBD.
- Supplementation with a specific species of this genus mitigates intestinal inflammation.
 - [Genomic and functional characterization of a *Butyricicoccus porcorum* strain isolated from human gut microbiota](#)
- Fits out hypothesis: During IL-6 elevation, the “good” microbiota are eliminated, leading to worse patient gut health.

Clostridia_vadinBB60_group

- Low abundance, poorly characterized, BUT preliminary data suggests it is overall beneficial to the host.
 - Has a protective effect (correlative) with respect to the development of Type 2 Diabetes Mellitus and coronary artery disease
 - [Altered Gut Microbiota as a Potential Risk Factor for Coronary Artery Disease in Diabetes: A Two-Sample Bi-Directional Mendelian Randomization Study](#)
 - Potential marker of the gut-brain axis health.
 - Persons with depression have lower levels of this compared to healthy controls.
 - Biomarker for microbiome based Alzheimer mice treatment.
 - Role in gut is unknown, but linked to increased dopamine metabolites or serotonin precursors.
 - Thought to potentially contribute to microbiome induced protection against experimentally induced colitis
 - Of the Clostridiales order, which is associated with reducing gut lumen inflammation by expressing Treg factors.

[Leveraging the microbiome to understand clinical heterogeneity in depression: findings from the T-RAD study](#)

Overall this also fits our hypothesis

Clostridium_sensu_stricto_1

- Mixed results on effects with respect to overall human health
 - Overgrowth is strongly associated with necrotic enteritis and reduced microbial diversity in small intestines of small animals
 - [Analysis of gut microbiota and the effect of lauric acid against necrotic enteritis in *Clostridium perfringens* and *Eimeria* side-by-side challenge model](#)
 - Elevated levels of this genus in small or proximal large intestine is a marker of dysbiosis, especially when it is accompanied by low *Lactobacillus* and other commensals. Damage is due to production of toxic products which damages mucosa when there is too much of this.

- [Gut Microbiota and Metabolite Changes in Patients With Ulcerative Colitis and Clostridioides difficile Infection](#)
- [Genus: Clostridium sensu stricto 1](#)
- May correlate with reduced cancer risk, in very specific contexts.

Prevotella

- Very diverse, well studied, context-dependent (either pro or against-health)
 - Prevotella abundance in the gut exists in 2 groups: some have high abundance, others have low abundance.
 - Longitudinal analysis shows Prevotella abundance is not constant: increases or decreases observed in samples from same person at specific time points suggests that Prevotella has transient dynamics (ie. Has huge abundance transiently, then returns to pre spike levels)
 - [Temporal dynamics and species-level complexity of Prevotella spp. in the human gut microbiota: implications for enterotypes and health](#)
 - Highly associated with high risk of IBS-D with visceral hypersensitivity (lowered pain tolerance in the lower gut extremities)
 - Enrichment co-occurs with Bacteroides, Bifidobacterium, Lactobacillus. Suspected dysbiosis tied to IBS-D instead of constipation predominated IBS.
 - [Altered Intestinal Microbiota with Increased Abundance of Prevotella Is Associated with High Risk of Diarrhea-Predominant Irritable Bowel Syndrome](#)
 - Suspected link between Prevotella overabundance and colitis, but not firmly established.
 - [Phylogenetic analysis of Prevotella copri from fecal and mucosal microbiota of IBS and IBD patients](#)

Muribaculaceae

- Enriched in populations with high fiber plant based diets, abundance is negatively correlated with high BMI (ie Muribaculaceae UP = BMI DOWN)
 - IBD is negatively correlated with Muribaculaceae, because it destroys the mucous layer of the intestinal tract, leaving Muribaculaceae fewer habitats.
 - Muribaculaceae is negatively correlated with type 2 diabetes, it has a synergistic role with diabetes medication as it changes the sugar metabolism.
 - [Exploration of the Muribaculaceae Family in the Gut Microbiota: Diversity, Metabolism, and Function](#)
 - Overall considered “good” for gut health